

Department of Pre Degree
UTMSPACE
Universiti Teknologi Malaysia

Test 1 (15%)

Time: 75 minutes

FSPM 0024 Calculus

Answer all questions.

1. Find the limit for each of the following.

a) $\lim_{x \rightarrow 0} \frac{\sqrt{3+x} - \sqrt{3}}{x}$. (3 marks)

b) $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x - 1}{7 - 2x^2}$. (3 marks)

2. Use First Principle Formula to find $f'(x)$ if $f(x) = 2x^2 + \frac{1}{x}$. (5 marks)

3. Use appropriate rules to find $\frac{dy}{dx}$ for each of the following.

a) $y = e^{x^2} \cos 3x$.
 $u = e^{x^2}, u' = 2e^{x^2}x$
 $v = \cos 3x, v' = -\sin 3x(3)$
 $\frac{dy}{dx} = uv' + u'v = e^{x^2}(-3\sin 3x) + 2e^{x^2}x(\cos 3x)$
 $= -3e^{x^2}\sin 3x + 2e^{x^2}x\cos 3x$ (3 marks)

b) $y = \ln(\sin^2 x)$.
 $u = \sin x, \frac{du}{dx} = \cos x$
 $y = \ln u^2$
 $\frac{dy}{dx} = \frac{2}{\sin x} \cos x$
 $= \frac{2\cos x}{\sin x}$
 $= 2\cot x$ (3 marks)

4. Given $y = \frac{x^6(1-x)^5}{1+x^3}$. By taking natural logarithm (ln) for both sides, express $\frac{dy}{dx}$ in terms of y and x . Hence, find the value of $\frac{dy}{dx}$ when $x = 2$. (4 marks)

5. Given the parametric equation

$$x = 4 - kt^2 \quad \text{and} \quad y = t^2 + 4t,$$

where k is a constant. Find

a) $\frac{dy}{dx}$ in terms of k and t , (2 marks)

b) the value of k if $\frac{d^2y}{dx^2} = 1$ when $t = -1$. (3 marks)

6. Find $\frac{dy}{dx}$ if $y = \frac{4}{x^2}$. Hence, use your answer to estimate the value of $\frac{4}{(2.98)^2}$. (4 marks)

Trigonometric	Differentiations
$\cos^2 x + \sin^2 x = 1$ $1 + \tan^2 x = \sec^2 x$ $\cot^2 x + 1 = \operatorname{cosec}^2 x$ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ $\sin 2x = 2 \sin x \cos x$ $\cos 2x = \cos^2 x - \sin^2 x$ $= 2 \cos^2 x - 1$ $= 1 - 2 \sin^2 x$ $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$	$\frac{d}{dx}[k] = 0, k \text{ constant.}$ $\frac{d}{dx}[x^n] = nx^{n-1}$ $\frac{d}{dx}[e^x] = e^x.$ $\frac{d}{dx}[\ln x] = \frac{1}{x}.$ $\frac{d}{dx}[\cos x] = -\sin x.$ $\frac{d}{dx}[\sin x] = \cos x.$ $\frac{d}{dx}[\tan x] = \sec^2 x.$ $\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x.$ $\frac{d}{dx}[\sec x] = \sec x \tan x.$ $\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x.$
Logarithm	
$a^x = e^{x \ln a}$ $\log_a x = \frac{\log_b x}{\log_b a}$	

$$1) \text{ (a)} \quad \lim_{x \rightarrow 0} \frac{\sqrt{3+x} - \sqrt{3}}{x} \times \frac{\sqrt{3+x} + \sqrt{3}}{\sqrt{3+x} + \sqrt{3}} = \lim_{x \rightarrow 0} \frac{3+x-3}{x(\sqrt{3+x} + \sqrt{3})}$$

$$= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{3+x} + \sqrt{3})} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{3+x} + \sqrt{3}} = \frac{1}{\sqrt{3} + \sqrt{3}} = \frac{1}{2\sqrt{3}}$$

$$\text{(b)} \quad \lim_{x \rightarrow \infty} \frac{3x^2 + 2x - 1}{7 - 2x^2} = \lim_{x \rightarrow \infty} \frac{\frac{3x^2}{x^2} + \frac{2x}{x^2} - \frac{1}{x^2}}{\frac{7}{x^2} - \frac{2x^2}{x^2}} = \lim_{x \rightarrow \infty} \frac{3 + \frac{2}{x} - \frac{1}{x^2}}{\frac{7}{x^2} - 2}$$

$$= \lim_{x \rightarrow \infty} \frac{3 + 0 + 0}{0 - 2} = -\frac{3}{2}$$

$$2) \quad f(x) = 2x^2 + \frac{1}{x}$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \frac{2(x+h)^2 + \frac{1}{x+h} - (2x^2 + \frac{1}{x})}{h}$$

$$= \frac{2x^2 + 4xh + 2h^2 + \frac{1}{x+h} - 2x^2 - \frac{1}{x}}{h}$$

$$= \frac{4xh + 2h^2 + \frac{x}{x+h} - \frac{1}{x^2+h}x}{h}$$

$$= 4x + 2h - \frac{\frac{x}{x+h} - \frac{1}{x}}{h}$$

$$= 4x + 0 + \frac{-1}{x^2+0}$$

$$= 4x - \frac{1}{x^2}$$

$$\frac{\frac{1}{x+h} - \frac{1}{x}}{x - (x+h)}$$

$$\frac{x - (x+h)}{x(x+h)}$$

$$4) y = \frac{x^6(1-x)^5}{1+x^3}$$

$$\ln y = \ln \left(\frac{x^6(1-x)^5}{1+x^3} \right)$$

$$\frac{1}{y} \frac{dy}{dx} = \ln x^6 + \ln(1-x)^5 - \ln(1+x^3)$$

$$\frac{1}{y} \frac{dy}{dx} = 6 \ln x + 5 \ln(1-x) - \ln(1+x^3)$$

$$\frac{1}{y} \frac{dy}{dx} = 6 \left(\frac{1}{x} \right) + 5 \left(\frac{-1}{1-x} \right) - \frac{3x^2}{1+x^3}$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{6}{x} + \frac{-5}{1-x} - \frac{3x^2}{1+x^3}$$

$$\frac{dy}{dx} = \frac{6y}{x} - \frac{5y}{1-x} - \frac{3x^2y}{1+x^3}$$

$$\text{When } x = 2, \frac{dy}{dx} = \frac{6y}{2} + \frac{-5y}{1-2} - \frac{3(2)^2y}{1+2^3}$$

$$= 3y + 5y - \frac{4}{3}y$$

$$= \frac{20}{3}y = \frac{20}{3} \left(-\frac{64}{9} \right) = -47 \frac{11}{27}$$

$$y = \cos^2(x^2-3)$$

$$y = \cos(x^2-3)^2$$

$$y = u^2$$

$$\frac{dy}{du} = 2u$$

$$u = \cos(x^2-3)$$

$$\frac{du}{dx} = -\sin(x^2-3)(2x)$$

$$= -2x \sin(x^2-3)$$

$$\frac{dy}{dx} = 2 \cos(x^2-3) (-2x) \sin(x^2-3)$$

$$= 2 \cos(x^2-3) (\sin x^2-3) (-2x)$$

$$= \sin(2(x^2-3)) (-2x)$$

$$= -2x \sin(2x^2-6)$$

$$x=2, y = -\frac{64}{9}$$

$$5) x = 4 - kt^2, y = t^2 + 4t, k \text{ is constant}$$

$$(a) \frac{dx}{dt} = -2kt \quad \frac{dy}{dt} = 2t + 4$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = 2t + 4 \left(\frac{1}{-2kt} \right) = \frac{2t+4}{-2kt} = \frac{t+2}{-kt}$$

$$(b) \frac{d^2y}{dx^2} = 1, t = -1$$

$$u = t+2, u' = 1$$

$$v = -kt, v' = -k$$

$$\frac{d^2y}{dx^2} = \frac{vu' - uv'}{v^2} \times \frac{dt}{dx}$$

$$= \frac{-kt(1) - (t+2)(-k)}{k^2t^2} \times \frac{1}{-2kt}$$

$$= \frac{-kt + kt + 2k}{k^2t^2} \times \frac{1}{-2kt}$$

$$= \frac{2}{k^2t^2} \times \frac{1}{-2kt}$$

$$= \frac{1}{-k^2t^3}$$

$$\frac{d^2y}{dx^2} = 1 = \frac{1}{-k^2t}$$

$$\frac{1}{-k^2(-1)} = 1$$

$$\frac{1}{k^2} = 1$$

$$k^2 = 1$$

$$k = 1$$

$$6) \quad y = \frac{4}{x^2} = 4x^{-2}$$

$$\frac{dy}{dx} = -8x^{-3} = -\frac{8}{x^3}$$

$$\text{Let } x_0 = 3, \quad y_0 = \frac{4}{9}, \quad \frac{dy}{dx} = -\frac{8}{27}$$

$$\Delta x = 2.98 - 3 = -0.02$$

$$\Delta y = \frac{dy}{dx} \cdot \Delta x$$

$$= -\frac{8}{27}(-0.02)$$

$$\approx \frac{4}{675}$$

$$\frac{4}{(2.98)^2} = y_N = y_0 + \Delta y$$

$$= \frac{4}{9} + \frac{4}{675}$$

$$= 0.4504$$

$$(a) y = e^{-x}$$

$$(b) y = e^{3x}$$

$$(c) y = e^{x^3}$$

$$(d) y = \sqrt{e^x}$$

$$(e) y = e^x + \frac{1}{e^{2x}}$$

$$(f) y = e^{\sqrt{2x+1}}$$

$$(g) y = e^{\sqrt{x}(x+3)}$$

$$(a) \frac{dy}{dx} = (e^{-x})(-1) = -e^{-x}$$

$$(b) \frac{dy}{dx} = e^{3x}(3) = 3e^x$$

$$(c) \frac{dy}{dx} = e^{x^3}(3x^2) = 3x^2e^{x^3}$$

$$(d) y = e^{x^{0.5}}$$

$$\frac{dy}{dx} = e^{x^{\frac{1}{2}}}(\frac{1}{2}x^{-\frac{1}{2}})$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{x}} (e^{x^{\frac{1}{2}}})$$

$$= \frac{1}{2\sqrt{x}} (\sqrt{e^x})$$

$$= \frac{\sqrt{e^x}}{2\sqrt{x}}$$

$$(e) \frac{dy}{dx} = e^x(1) + e^{-2x}(-2)$$

$$= e^x - \frac{2}{e^{2x}}$$

$$(f) \frac{dy}{dx} = e^{(\sqrt{2x+1})^{\frac{1}{2}}} \left(\frac{1}{2}(\sqrt{2x+1})^{-\frac{1}{2}} \times \frac{1}{\sqrt{2x+1}} \right)$$

$$= e^{\sqrt{2x+1}} \times \frac{1}{\sqrt{2x+1}} = \frac{e^{\sqrt{2x+1}}}{\sqrt{2x+1}}$$

$$(g) \frac{dy}{dx} = e^{\sqrt{x}(x+3)} \left(\frac{3x+3}{2\sqrt{x}} \right)$$

$$x^{\frac{3}{2}} + 3x^{\frac{1}{2}}$$

$$= \frac{3}{2}x^{\frac{1}{2}} + \frac{3}{2}x^{-\frac{1}{2}}$$

$$= \frac{3}{2}\sqrt{x} + \frac{3}{2}\frac{1}{\sqrt{x}}$$

$$= \frac{3\sqrt{x}}{2} + \frac{3}{2\sqrt{x}}$$

$$= \frac{3x+3}{2\sqrt{x}}$$

$$(i) y = 2x^2e^x, \frac{dy}{dx} = 4xe^x(1) = 4xe^x$$

$$(ii) y = \sqrt{3-2e^{3x}}$$

$$y = u^{\frac{1}{2}} \quad \frac{dy}{dx} = \frac{1}{\sqrt{u}} \quad \frac{dy}{dx} = \frac{1}{\sqrt{3-2e^{3x}}}$$

$$u = 3-2e^{3x} \quad \frac{du}{dx} = 0 - 2e^{3x}(3)$$

$$= -6e^{3x}$$

$$(iii) y = (e^x - e^{-x})^7$$

$$y = u^7$$

$$\frac{dy}{du} = 7u^6 \quad \checkmark$$

$$u = e^x - e^{-x}$$

$$\frac{du}{dx} = e^x + e^{-x} \quad \checkmark$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = 7(e^x - \frac{1}{e^x})^6 (e^x + \frac{1}{e^x})$$

$$(iv) y = \frac{e^{2x}}{e^x + e^{-x}}$$

$$u = e^{3x}, v = e^x + e^{-x}$$

$$u' = 3e^{3x}, v' = e^x - \frac{1}{e^x}$$

$$\frac{dy}{dx} = \frac{uv' - u'v}{v^2}$$

$$= \frac{e^x + \frac{1}{e^x}(3e^{3x}) - e^{3x}(e^x - \frac{1}{e^x})}{(e^x + \frac{1}{e^x})^2}$$

$$= \frac{3e^{4x} + 3e^{2x} - e^{4x} + e^{2x}}{(e^x + e^{-x})^2}$$

$$= \frac{2e^{4x} + 4e^{2x}}{(e^x + e^{-x})^2}$$

$$(c) \quad y = \ln(3x-1), \quad u = 3x-1$$

$$\frac{dy}{dx} = \frac{y'}{u} = \frac{3}{3x-1}$$

$$y = \ln \frac{\sqrt{x-1}}{\sqrt{x+1}} = \frac{1}{2} \ln \left(\frac{x-1}{x+1} \right)$$

$$\frac{dy}{dx} = \frac{1}{2} \left(\frac{y'}{u} \right) = \frac{1}{2} \left(\frac{\frac{2}{(x+1)^2}}{\frac{x-1}{x+1}} \right) = \frac{1}{2} \left(\frac{2}{(x+1)(x-1)} \right)$$

$$= \frac{1}{(x+1)(x-1)}$$

$$y = \ln \sqrt{\frac{x^2+4x+1}{x-3}}$$

$$y = \frac{1}{2} \ln \frac{x^2+4x+1}{x-3}$$

$$u = x^2+4x+1 \quad u' = 2x+4$$

$$v = x-3 \quad v' = 1$$

$$y = \frac{1}{2} \left(\frac{\frac{x-3(2x+4) - (x^2+4x+1)}{(x-3)^2}}{\frac{x^2+4x+1}{(x-3)}} \right)$$

$$= \frac{1}{2} \left(\frac{2x^2-2x-12-x^2-4x-1}{(x-3)(x^2+4x+1)} \right)$$

$$= \frac{x^2-6x-13}{2(x-3)(x^2+4x+1)} = \frac{x^2-6x-13}{2x^3+2x^2-22x-6}$$

$$y = \cos(2x+\pi)$$

$$\frac{dy}{dx} = -2\sin(2x+\pi)$$

$$y = \tan \frac{\pi}{4} x$$

$$\frac{dy}{dx} = \sec^2 \left(\frac{\pi x}{4} \right) \left(\frac{\pi}{4} \right)$$

$$y = \sec 2x$$

$$\frac{dy}{dx} = 2\sec 2x \tan 2x$$

$$y = \operatorname{cosec} \left(\frac{\pi+1}{2} \right) x$$

$$\frac{dy}{dx} = -\operatorname{cosec} \left(\frac{\pi+1}{2} \right) \cot \left(\frac{\pi+1}{2} \right) \left(\frac{\pi+1}{2} \right)$$

$$y = \cot(2x^2-1)$$

$$\frac{dy}{dx} = -4x \operatorname{cosec}^2(2x^2-1)$$